

1. Administrative & Study Identification

Full Study Title: A Randomized, Placebo-controlled, Parallel Group, 72-week Study to Evaluate the Efficacy and Safety of VHB937 in Participants With Early Alzheimer's Disease Followed by an Extension **Short Title/Acronym:** A Clinical Trial to Learn About the Effects of VHB937 in People With Early Alzheimer's Disease **Protocol Number:** CVHB937A12201 **NCT Number:** NCT07094516 **Study Status:** Recruiting **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** VHB937 (TREM2 stabilizing and activating fully human monoclonal antibody) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations / Ana Graf (Senior Global Program Head)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of VHB937 in participants with early Alzheimer's disease (AD), measured by the change in the Clinical Dementia Rating-Sum of Boxes (CDR-SB) scale over a 72-week treatment period. **Secondary Objectives:** To evaluate the drug's effect on secondary cognitive and functional scales including the Alzheimer's Disease Assessment Subscale 14 (ADAS-Cog14) and the Alzheimer's Disease Cooperative Study-Activities of Daily Living (ADCS-ADL), as well as to assess safety, tolerability, pharmacokinetics/pharmacodynamics, and immunogenicity.

2.2 Background & Rationale To address the progression of cognitive and functional decline in patients with early Alzheimer's Disease. Stabilizing and activating TREM2, a receptor found on microglia, may help protect nerve cells and influence the actions of brain immune cells. By doing so, VHB937 aims to slow Alzheimer's disease progression and prevent further nerve cell damage.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized, Parallel Group (1:1:1 to VHB937 Low Dose, VHB937 High Dose, or Placebo)

3.2 Planned Enrollment & Duration Target **NS:** 407 evaluable participants **Number of Sites:** Multicentre (Global) **Estimated Study Start:** 07-Aug-2025 **Estimated Primary Completion:** 14-Sep-2028 **Estimated Study Completion:** 31-Dec-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 50 to 85 years old (Sex: All). 2. Diagnosis of Mild Cognitive Impairment (MCI) due to Alzheimer's disease or mild Alzheimer's disease according to the NIA-AA criteria. 3. Clinical Dementia Rating (CDR) Global score of 0.5 or 1.0. 4. Confirmation of Alzheimer's disease based on cerebral spinal fluid (CSF) biomarkers or amyloid PET imaging. 5. Must have a reliable study partner to accompany them to visits.

4.2 Exclusion Criteria 1. Lack of a reliable study partner. 2. Unstable doses of current AD treatments (participants currently receiving symptomatic AD treatments such as acetylcholinesterase inhibitors or memantine must be on a stable dose before starting the trial). 3. Any unstable medical condition or finding that would compromise the participant's safety or study outcomes.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: VHB937 Low Dose or VHB937 High Dose versus Placebo. **Route of Administration:** Intravenous (I.V.) infusions. **Duration of Treatment:** 72 weeks of double-blind treatment, followed by an open-label extension.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of concomitant symptomatic Alzheimer's disease medications (e.g., acetylcholinesterase inhibitors, memantine). **Prohibited:** Other investigational agents during the duration of the trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	2-4 Weeks	Screening Informed Consent, Medical History, CSF/PET Biomarker confirmation, baseline cognitive assessments
2	Day 0	Baseline Randomization, First I.V. Infusion, Baseline Safety Labs
3-72	Over 72 Weeks	Interim Regular visits for I.V. infusions, Safety Labs, Efficacy Assessments (CDR-SB, ADAS-Cog14), PK/PD sampling
73	Week 72	End of Study Final Efficacy Assessment, Final Vital Signs, Safety Follow-Up, Transition evaluation to Open-Label Extension

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in the Clinical Dementia Rating-Sum of Boxes (CDR-SB) scale at 72 weeks. **Measurement Method:** Clinical assessment measuring cognition and function across six domains: Memory, Orientation, Judgment &

Problem Solving, Community Affairs, Home & Hobbies, and Personal Care.

7.2 Secondary & Exploratory Endpoints - Change from baseline on the Alzheimer's Disease Assessment Subscale 14 (ADAS-Cog14). - Change from baseline on the Integrated Activities of Daily Living domain on the Alzheimer's Disease Cooperative Study-Activities of Daily Living (ADCS-ADL). - Incidence and severity of Adverse Events (AEs) and Serious Adverse Events (SAEs). - Pharmacokinetics/pharmacodynamics and immunogenicity.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of incidence and severity of AEs, laboratory tests, vital signs, and ECG findings. **Serious Adverse Events (SAEs):** Monitored and reported by investigators to the study sponsor within standard reporting timelines. **Data Monitoring Committee (DMC):** Regular safety reviews conducted to ensure participant safety across the multicentre trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary cognitive endpoints; Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** Powered with an estimated 407 participants to detect statistically meaningful slowing of decline in CDR-SB over 72 weeks between VHB937 and Placebo groups. **Handling of Missing Data:** Standard longitudinal multiple imputation and mixed-model repeated measures (MMRM) for cognitive decline assessments.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite and centralized remote monitoring visits to ensure integrity of intravenous infusion administration and cognitive test administration. **Audit Trail:** Validated eCRF tracking and central rater review for primary cognitive assessments (CDR-SB).

11. Study Identifiers & Governance

Primary ID: CVHB937A12201 **Registry Number:** NCT07094516 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** A Clinical Trial to Learn About the Effects of VHB937 in People With Early Alzheimer's Disease **Official Regulatory Title:** A

Randomized, Placebo-controlled, Parallel Group, 72-week Study to Evaluate the Efficacy and Safety of VHB937 in Participants With Early Alzheimer's Disease Followed by an Extension

12. Clinical Framework (Alzheimer's Disease Specific)

Primary Condition: Early Alzheimer's Disease (MCI due to AD or mild AD) **Diagnosis Threshold:** CDR Global score of 0.5 or 1.0 with biomarker confirmation (CSF or amyloid PET) **Medical Methodology:** TREM2 stabilization and activation via monoclonal antibody **Optimization Target:** Protection of nerve cells and slowing of Alzheimer's disease progression

13. Study Procedures & Methodology

Management Pathway: Standard Alzheimer's clinical evaluation combined with periodic I.V. infusions **Education Protocol (HCP):** Site initiation and rater training for cognitive assessments (CDR-SB, ADAS-Cog14) **Education Protocol (Patient):** Mandatory reliable study partner/caregiver inclusion to assist with accurate functional reporting **Quality Audit Mechanism:** Centralized rater monitoring and AE tracking to ensure cognitive test reliability

14. Observation & Follow-Up Schedule

Total Duration: 72 weeks of double-blind treatment followed by long-term open-label extension **Onsite Visit Cadence:** Regular scheduled visits for I.V. infusions and cognitive evaluation battery **Remote Monitoring Frequency:** Intermittent safety follow-up checks between infusion visits **Checklist Review Interval:** Cognitive/functional scale evaluations scheduled at baseline and prespecified interim points up to week 72

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				